

Smaran Vallabhaneni

(617) 366-6779 | smaranvallabhaneni@gmail.com | [linkedin.com/in/smaran-vallabhaneni](https://www.linkedin.com/in/smaran-vallabhaneni) | github.com/happysmaran | smaran-vallabhaneni.com

EDUCATION

Olin College of EngineeringNeedham, MA

Bachelor of Science, Engineering in Computing (GPA: 4.0)Aug. 2025 – May 2029

– **Relevant Courses:** Modeling & Simulation, Linear Algebra, Data Science, Multi-Variable Calculus, Software Design (Python), Kernel Development (C), Cryptography, Algorithms and Data Structures, Machine Learning, Products & Markets, Introduction to Sensors, Instrumentation & Measurement

SKILLS

Programming Languages: Java, C/C++, Python, pandas, HTML, CSS, JavaScript, TypeScript, Dart, LaTeX, MySQL, NoSQL, C#, R, PBASIC, Bash, Zsh, PowerShell, Swift, MATLAB

Operating Systems / SBCs: Windows, Unix, macOS, Linux (Raspberry Pi OS, Debian, Fedora, and Arch based), Raspberry Pi, Arduino

Tools: Blender, Fusion360, AutoCAD, Autodesk, SolidWorks, Unity, Figma, Adobe Creative Suite, Excel, Git, IntelliJ, Eclipse IDE, VSCode, Xcode, Overleaf, Node.js, Next.js, GPL Cver, Simulink, MATLAB, Tableau Desktop, Docker, MissionPlanner, ArduPilot, GitHub Copilot, Google Gemini, Claude Code, LM Studio

Frameworks: LLMs, RAG Systems, Django, PyTorch, NumPy, Tkinter, Scikit-Learn, Kivy, Flutter, SwiftUI, React, AngularJS, .NET, Ollama, System Design, Metal API, ROS/MAVROS, OpenCV, MediaPipe

Other: UI/UX, CI/CD Pipelines, ESP32, Computer Vision, Internet of Things, AWS, GCP, Azure, Dev Ops, HuggingFace, NLP, Databases, Database Management, Statistical Analysis, Version Control, Embedded Systems, Algorithm Development, Data Structures, Object-Oriented Programming, API Integration, Testing and Debugging

Soft Skills: Communication, Problem Solving, Team Collaboration, Leadership, Adaptability, Time Management, Attention to Detail

PROJECTS

CLOS (Command Line Operating System) | C++, Java, HTML, CSS, BashApr. 2023 – Present

– Building a cross-platform centralized terminal and graphical math tool with equation solving, graphing, symbolic calculus, trigonometry, and more.

– Engineered a custom database engine, symbolic math parser, custom scripting language, and a 3D rendering library from scratch in Java; achieved consistent 60 FPS graphic rendering and executed complex multi-variable calculus computations under 50 ms.

ClassicTunes | Swift, SwiftUI, Metal APIOct. 2025 – Jul. 2026

– Developed an open-source modern remake of iTunes 7–10 for Apple Silicon macOS, faithfully recreating CoverFlow, MiniPlayer, the iTunes Store, iPod Syncing, and Smart/Genius Playlists.

– Leveraged Apple’s Metal API to build a custom GPU-accelerated CoverFlow engine, matching the performance of the native Apple Music client with a fluid 120 FPS render rate across 150+ albums on Apple Silicon.

Whoa-Scope | Python, KivyFeb. 2026 – Apr. 2026

– Co-developed an open-source oscilloscope fork to resolve compatibility issues and desktop UX flaws, gaining organic adoption from 50% of the class (40 students) and faculty recognition.

– Profiled and refactored memory allocation, slashing RAM consumption by 200+ MB to maximize stability and prevent lag on lab computers.

Specter | Python, Linux, LLMsMay 2026

– Created a local-first, privacy oriented AI screen agent for Fedora Wayland to perform tasks efficiently with simple prompts.

– Locally ran with a response time of 3 seconds per action, with extreme fidelity and understanding of on-screen content on a capable GPU.

3D Gesture Tracking | Blender, Python, OpenCVApr. 2026

– Engineered a real-time computer vision add-on, enabling mouse-free viewport manipulation (pan, zoom, rotate) via hand gesture tracking.

W.A.G.E | Python, TypeScript, Next.js, ReactJul. 2026

– Built a local-first AI job search strategy tool designed to make pre-application strategies more streamlined, with a fit score, tailored resume bullets, and outreach drafting. Designed with ease of use in mind, featuring full job importing and management in a slick user interface.

EXPERIENCE

Olin College AERO Project TeamNeedham, MA

Head of Ground Station Development — Software EngineerSep. 2025 – Present

– Leading software development for an autonomous fixed-wing aircraft competing in the AUVSI SUAS competition, targeting precision water delivery to GPS-specified targets using image recognition.

– Architected ground station software integrating ArduPilot, MissionPlanner, and ROS/MAVROS systems to bridge an on-board NVIDIA Jetson computer with ground controls.

– Engineered a robust fly-by-wire telemetry pipeline, achieving a consistent sub-100ms round-trip latency for real-time sensor monitoring and safety interlocks.

Geeky InsightsRemote

Full-Stack Android App Developer — InternshipApr. 2024 – Apr. 2025

– Designed end-to-end UI/UX of a customizable audio alarm clock app using Figma and Adobe Creative Suite, prioritizing accessibility and intuitive design for a diverse user base.

– Engineered an end-to-end frontend/backend architecture using Flutter, Dart, and Java, optimizing the compilation pipeline into a lightweight, bug-free 15 MB production package.

– Optimized application memory layout to support low-resource devices running Android 9+, reducing RAM consumption to match the baseline footprint of Android’s native system clock.

ACTIVITIES

Olin ICPCNeedham, MA

Competitive Programmer — Team MemberJan. 2026 – Present

– Solving high-difficulty algorithmic problems under competitive time constraints in bi-weekly ICPC-style contests.

– Applying advanced techniques including segment/Fenwick trees, dynamic programming, graph algorithms, string algorithms, network flow, and computational geometry.